INCH-POUND
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MILITARY SPECIFICATION

TESTER, DIESEL FUEL INJECTOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of manually operated tester for diesel fuel injectors.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation (see 6.2.1).

SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Box, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Shipping, Fiberboard.
PPP-B-640	-	Box, Fiberboard, Corrugated, Triple-Wall.
PPP-C-843	-	Cushioning Material, Cellulosic.
PPP-F-320	-	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-T-60	-	Tape, Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-ESM-P, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4910

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MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods.
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the DODSSP, Standardization Document Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

US DEPARTMENT OF LABOR

29 CFR 1910 - General Industry, OSHA Safety and Health Standards Chapter XVII, Part 1910

(Unless otherwise indicated, copies of the regulation are available from the Superintendent of Documents, Government Printing Office, Washington DC 20402.)

DEPARTMENT OF ARMY

DRAWING

7551255 - Test Set Fuel Injection

(Copies of specifications, standards, and other Government drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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* 2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issue of the document cited in the solicitation (see 6.2.1).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951 - Standard Practice for Commercial Packaging.
ASTM E380 - Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2.1) in the contract or purchase order, a sample shall be subjected to first article inspection (see 6.2.1) in accordance with 4.3.

3.2 Design. The tester shall be new and of the manufacturer's latest design that shall meet the performance requirements specified herein. The tester shall be designed for manual operation and portability. The design shall assure that the tester be capable of testing diesel engine injectors for the following:

(a) Opening or popping pressure.
(b) Spray pattern.
(c) Nozzle and orifice leakage.
(d) Assembly joint leakage.
(e) Back pressure check valve functioning.
(f) Unit injection nozzle metering.

The design shall be such that parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, or repair without modification, distortion, or damage to the tester or any tester part or component.

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3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the tester. In this specification, most measurements, dimensions, sizes, and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380.

3.2.2 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.2.3 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

* 3.2.4 Safety and health requirements. The tester shall comply with the general safety and health requirements promulgated under 29 CFR, Chapter XVII, Part 1910 that are applicable to the tester itself. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.4).

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Construction. The tester shall be furnished complete so it can be used for the operations specified herein. Parts used in fabricating the tester shall be new and free of defects. All part surfaces shall be clean and free of extraneous materials. External surfaces shall be smooth and free of sharp edges. Functional surfaces shall be free from burrs and defects affecting performance.

3.4.1 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when subject to renewal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.5 Performance and product characterisitics.

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* 3.5.1 Pump capacity. The pump shall be capable of developing up to 4,500 pound-force per square inch gage (psig) pressure, and of delivering test fluid within a minimum range of not less than 600 to not more than 1,400 cubic millimeters per stroke.

3.5.2 Gage accuracy. The gage accuracy in the middle half of the scale shall be within two percent of the maximum pressure for which the scale is graduated. Accuracy for the balance of the scale shall be within three percent of the maximum pressure for which the scale is graduated.

3.5.3 Pressure retention. The tester, with all outlets capped, shall be capable of containing a pressure of 3,000 psig for a period of one minute with an allowable pressure drop not exceeding 100 psig and with no evidence of external fluid leakage.

3.6 Components.

3.6.1 Pump. The pump shall be a hand operated, self-priming, high pressure type constructed so that it can be bolted or clamped to a bench, mounted on a base plate, or rested on a flat surface.

3.6.1.1 Fluid reservoir. The test fluid reservoir shall have a minimum capacity of two-thirds of a pint. The reservoir, including cover, shall be splash-proof and shall be located to permit easy filling with test fluid.

3.6.1.2 Filter. The filter shall be rated at five microns, and shall be of a replaceable element type.

3.6.1.3 Pressure gage. The pressure gage shall have a minimum range of 0 to 5,000 pound-force per square inch gage (psig). The gage dial shall be divided into increments not greater than 50 psig, with numerical markings at least every 500 psig. The gage dial shall have a minimum diameter of 3.5 inches. The dial shall be either black with white graduations and numerals, or white with black graduations and numerals.

3.6.1.4 Valve system. The valve system of the pump shall include means to isolate the gage during rapid, prolonged pumping when flushing the nozzle or when conducting spray pattern testing, and to isolate the pump during nozzle leakage testing.

3.6.2 Accessories.

3.6.2.1 Holding fixtures. When required (see 6.2.1), injector holding fixtures, base plates, actuators, etc., shall be provided for testing hydraulically and mechanically actuated injectors as specified by the procuring activity. Fixtures shall be designed to rest freely on a flat surface but shall incorporate means to allow bolting or clamping to a bench.

3.6.2.2 Collector cup. A collector cup and a graduated measuring container or a graduated transparent collector cup, shall be provided for fluid discharge and leakage tests. The collector cup shall have a capacity of at least 250 cubic centimeters.

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3.6.2.3 Connector tube or hose. A flexible, high pressure tube or hose not less than 24 inches in length shall be furnished with each tester. The hose or tubing shall be capable of connecting the tester to the nozzles which are to be tested. Each end of the tubing or hose shall terminate in a 90 degree angle fitting with a 60 degree male taper and 1/2-20 National Fine left hand male threads. The tubing or hose assembly shall be compatible with the functional operating conditions and pressures encountered during use.

* 3.6.2.4 Connector set. Connector nuts, tubes, adapters, and fittings needed for making connections between the tester and the injectors to be tested shall be supplied as specified per drawing 7551255.

3.6.3 Carrying case. A carrying case shall be provided with each tester. The case shall be capable of holding the complete tester with all required accessories and technical data securely in place. The carrying case shall be fabricated of sheet metal with a thickness not less than 0.032 inch. The case shall include a carrying handle that assures a comfortable four finger grip. The cover of the case shall be hinged and secured by snap-catch locks. The total weight of the case, including contents, shall not exceed 70 pounds.

3.7 Painting. Painting and finishing shall be in accordance with the best practice of the industry. Immediately prior to painting, surfaces shall be clean, dry, and free of foreign matter. The protective paint coating shall be not less than 2.0 mil in thickness. The paint adhesion shall be such that no paint shall be peeled by fiberglass tape, when two intersecting knife cut expose the base metal for a distance of not less than two inches. The surfa hardness of the paint shall be between the industrial standards of H to HB for pencil lead.

3.8 Product identification. The tester shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.9 Workmanship. Standards of workmanship shall assure that the tester shall have the stability, durability, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. When specified (see 6.2.1), first article inspection shall be applied to a preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the product examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (f) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Pressure gage accuracy (see 4.6.1).
(c) Pump priming and delivery test (see 4.6.2).
(d) Pressure retention test (see 4.6.3).
(e) Paint test (see 4.6.4).
(f) Packaging inspection (see 4.7).

* 4.4.1 Sampling. Sampling for quality conformance inspection shall be performed as listed with all samples selected randomly.

Inspection or test	Lot or batch size	Sample size
Product examination (see 4.5).	100 percent	-
Pressure gage accuracy (see 4.6.1).	1-150	13
Pump priming and delivery (see 4.6.2).	1-150	13
Pressure retention test (see 4.6.3).	1-150	13
Paint test (see 4.6.4).	1-150	3
Packaging inspection (see 4.7).	1-150	5

The lots shall not exceed the maximum sizes indicated above. If lot size is less than or equal to sample size, 100 percent inspection is required. Each lot shall be accepted with no defects and rejected if one or more defects are found.

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4.5 Product Examination. Visually, dimensionally, and manually examine each tester to determine conformance with the requirements of 3.2 through 3.4.1 and 3.6 through 3.9. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Dimensional examinations includes measuring dimensions as specified. Manual examination shall include the operation of moveable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacturing at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the tester and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.2.3), shall be cause for rejection.

4.6 Tests.

4.6.1 Pressure gage accuracy. The pressure gage shall be removed from the tester and tested for accuracy by comparison with a standard hydraulic pressure measuring device having an over-all accuracy within one-fourth of one percent (see 3.5.2).

4.6.2 Pump priming and delivery test. The test fluid reservoir shall be filled with a diesel calibrating oil. The collector cup shall then be held under the fluid outlet and the pump handle shall be actuated ten full strokes. The amount of fluid resulting from the ten strokes shall be measured to determine conformance to 3.5.1. This test shall be performed at a temperature of +70 degrees ±10 degrees Fahrenheit.

4.6.3 Pressure retention test. The fluid outlets shall be capped or plugged, and the pump shall be actuated until the gage registers a pressure of not less than 3,000 psig. The operator shall then let the pump stand inactive for a period of one minute. At the end of the one minute period, the pressure registered on the gage shall be noted for compliance with 3.5.3, and the unit examined for evidence of external leakage.

4.6.4 Paint test. The hardness, adhesion and thickness of the paint applied to the tester shall be checked as follows:

(a) The graphite point of a sharpened HB lead pencil shall be pressed at a right angle into the painted surface of the tester frame until the graphite crumbles. Failure of the surface to withstand the test, as evidenced by a pitting of the paint, shall be cause for rejection.
(b) A test sample of the same metal, prepared and painted with the same process and at the same time as the tester, shall have a sharpened H lead pencil pressed into the paint as in (a). Failure of the surface to pit (paint too hard) shall be cause for rejection.

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(c) Two 2 inch intersecting lines shall be cut through the paint to the base metal of the sample. A strip of 1/2 inch wide (minimum) fiberglass tape shall be firmly pressed over the intersection and then pulled from the surface. Adhesion of any paint to the tape shall be cause for rejection.

(d) Using a paint thickness measuring instrument, or by masking a portion of the test sample during painting, measure the thickness of the paint. Failure of the paint to be at least 0.0020 inch thick shall be cause for rejection.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

* 5.1 Preservation. Preservation shall be Level A or Level C or ASTM D3951 as specified in the contract (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The tester and components shall be cleaned and dried by any applicable process and procedure of MIL-P-116.

5.1.1.2 Preservation application. Preservatives and procedures for application shall conform to MIL-P-116. Immediately after cleaning and drying, flush the interior surfaces of the fluid system with a preservative oil conforming to type P-10, grade 30. The main plunger shall then be actuated to assure a protective preservative oil coating around the plunger seal to prevent seizure within the chamber. All exterior, ferrous metal surfaces of the tester and accessories, which have not been painted or afforded corrosion protection, shall be coated with type P-7 preservative. Excess preservative shall be allowed to drain.

5.1.1.3 Unit packaging. The complete tester shall be packaged in accordance with method I of MIL-P-116. Wrap all exterior preserved surfaces with barrier material conforming to MIL-B-121, grade A. Wrap the pressure gage with one inch thick, uncompressed, cushioning material conforming to PPP-C-843, type II. Secure cushioning and barrier material in place with tape conforming to PPP-T-60, type optional. Close all apertures of the tester with plastic caps or plugs conforming to MIL-C-5501. Similar type plastic caps or plugs shall be placed over all exposed, unprotected threaded surfaces. The caps or plugs shall be snug-fitting or shall be taped in place with tape of PPP-T-60, type optional. Place all components in their respective compartments or stowage brackets of the carrying case and secure with means provided. Fiberboard pads or cut shapes conforming to PPP-F-320, class domestic, shall be employed inside the carrying case to restrict parts which may become loosened from their stowage brackets during shipment. Technical data, when applicable, shall be unit packaged in accordance with method IC-1 of MIL-P-116 and placed inside the carrying case. The carrying case shall then be placed in a fiberboard container conforming to PPP-B-636, type CF, class weather resistant. The unit container shall be waterproof sealed in accordance with the appendix to PPP-B-636.

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* 5.1.2 Level C. Level C preservation shall be as specified in MIL-STD-1190. In addition, immediately after cleaning and drying, flush the interior surfaces of the fluid system with a preservative oil. The main plunger shall then be actuated to assure a protective preservative oil coating around the plunger seal to prevent seizure within the chamber.

* 5.1.3 Commercial. Commercial preservation shall be as specified in ASTM D3951.

* 5.2 Packing. Packing shall be Level A, Level B, or Level C or ASTM D3951, as specified in the contract (see 6.2.1).

5.2.1 Level A. A quantity of testers, unit packaged as specified in 5.1.1.3, shall be packed in a box conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. The gross weight of the exterior shipping container shall be governed by the limitations of the container specification. Blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186. Closure and strapping shall be in accordance with the applicable container specification and the appendix thereto.

5.2.2 Level B. Testers shall be packed as specified in 5.2.1 except the shipping containers shall conform to PPP-B-601, domestic type, PPP-B-621, class 1, or PPP-B-640, class 2.

* 5.2.3 Level C. Level C packing shall be as specified in MIL-STD-1190.

* 5.2.4 Commercial. Commercial packing shall be as specified in ASTM D3951.

* 5.3 Marking. All marking shall be in accordance with MIL-STD-129, or ASTM D3951 as specified in the contract (see 6.2.1).

6. NOTES

(This section contains information of a general explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The injector tester covered by this specification is used to check opening or popping pressure, assembly joint leakage, spray pattern, nozzle and orifice leakage, back pressure check valve functioning, and metering by unit injection nozzles of diesel engines.

6.2 Ordering data.

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* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.)
(c) First article when required (see 3.1).
(d) Exceptions and additional safety/health requirements when required (see 3.2.4).
(e) Holding fixtures required (see 3.6.2.1).
(f) Product identification if different (see 3.8).
(g) First article inspection when required (see 4.3).
(h) Quality conformance inspection if different (see 4.4).
(i) Level of preservation, packing and marking required (see 5.1, 5.2 and 5.3).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 Subject term (key word) listing.

Injector
Nozzle
Orifice
Leakage

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Custodians:

Army - AL
Navy - YD
Air Force - 99

Review activities:

Army - AT
Navy - MC, SH
Air Force - 82
DLA - CS

User activities:

Army - ME
Navy - OS

Preparing activity:

Army - AL

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